

Data Analytics (DA)

Module -1 - Excel

Excel Assignment 2 – Data Exploration

Assignment Task Overview

Problem Statement

As a **Data Analyst**, you are required to perform **data cleaning and preparation using Microsoft Excel**. Real-world datasets often contain **missing values, inconsistent formatting, duplicate records, and poorly structured text fields**, which must be cleaned before performing meaningful analysis.

In this assignment, you will apply Excel's **data cleaning techniques, formatting tools, and transformation operations** to prepare the dataset for analysis. The goal is to develop practical skills in **data preprocessing, standardization, and quality improvement**.

The focus of this assignment is on **handling missing values, correcting inconsistencies, removing duplicate records, restructuring columns, and applying formatting rules** to improve the usability and readability of the dataset.

Dataset Description

Dataset: [Product Dataset](#)

Attributes: *Product ID, Product Name, Brand Name, Quantity, Category, Price*

Tasks:

1. Handling Missing Values:

- Check for missing values in the 'Price' column. How would you handle products with missing price information?
- If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

2. Correcting Inconsistent Data:

- Identify any inconsistent text formats present in the "Product Name" column.
- Identify any typos present in the "Category" column.
- Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.

3. Removing Duplicates:

- Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

4. Splitting and Merging Data:

- Split the "Product ID" column into two separate columns for "Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.
- Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".

5. Number Formatting:

- Format the data type of the "Price" column to a currency format.
- Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY" format.

6. Conditional Formatting:

- Apply data bar or color scales conditional formatting in the "Price" column.
- Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."

Deliverables:

Submit a project folder containing:

1. Excel File (.xlsx)

The Excel workbook should include:

- Cleaned dataset
- Corrected text formats
- Removed duplicates
- Split and merged columns
- Applied number formatting
- Conditional formatting rules

2. PDF Document

Include screenshots of:

- Missing value handling
- Find and Replace corrections
- Duplicate removal process
- Column splitting and merging
- Conditional formatting results

Provide a brief explanation for each cleaning step.

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
Missing Value Handling	Data Cleaning	Handling Null Values	Beginner
Data Standardization	Data Preparation	Text Standardization	Beginner
Duplicate Handling	Data Quality	Removing Duplicate Records	Beginner
Data Transformation	Data Manipulation	Splitting & Merging Columns	Beginner
Data Formatting	Data Presentation	Number & Date Formatting	Beginner
Conditional Formatting	Data Visualization	Highlighting Patterns	Beginner

Rubric & Mark Allocation (Total 10 Marks)

Skill / Component	Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
Missing Value Handling	2	Correct identification and handling strategy	Partial handling strategy	1
Data Standardization	2	Correct correction of text formats and typos	Minor inconsistencies	1
Duplicate Removal	2	Correct identification and removal of duplicates	Some duplicates remain	1
Data Transformation	2	Correct splitting and merging of columns	Partial extraction	1
Formatting & Visualization	2	Proper number formatting and conditional formatting	Incorrect formatting	1

DA Skill Taxonomy Structure

- **Data Cleaning:** Handling missing values and correcting inconsistencies
- **Data Quality Management:** Removing duplicate records
- **Data Transformation:** Splitting and merging columns
- **Data Formatting:** Applying number and date formats
- **Visual Highlighting:** Using conditional formatting for insights
- **Spreadsheet Proficiency:** Using Excel tools for data preparation

Evaluation Guidelines

- Verify that missing values are properly handled.
- Ensure text formats and category names are standardized.
- Confirm duplicate records have been removed.
- Validate column splitting and merging operations.
- Check formatting and conditional formatting rules.
- Assign marks based on accuracy, completeness, and clarity of the cleaned dataset.

Achievement Criteria & Passing Rules

- Students must obtain partial marks in every assignment section.
- All cleaning operations must be correctly applied and documented.
- The dataset must be fully cleaned and formatted for analysis.
- Screenshots and explanations must clearly demonstrate the steps performed.